

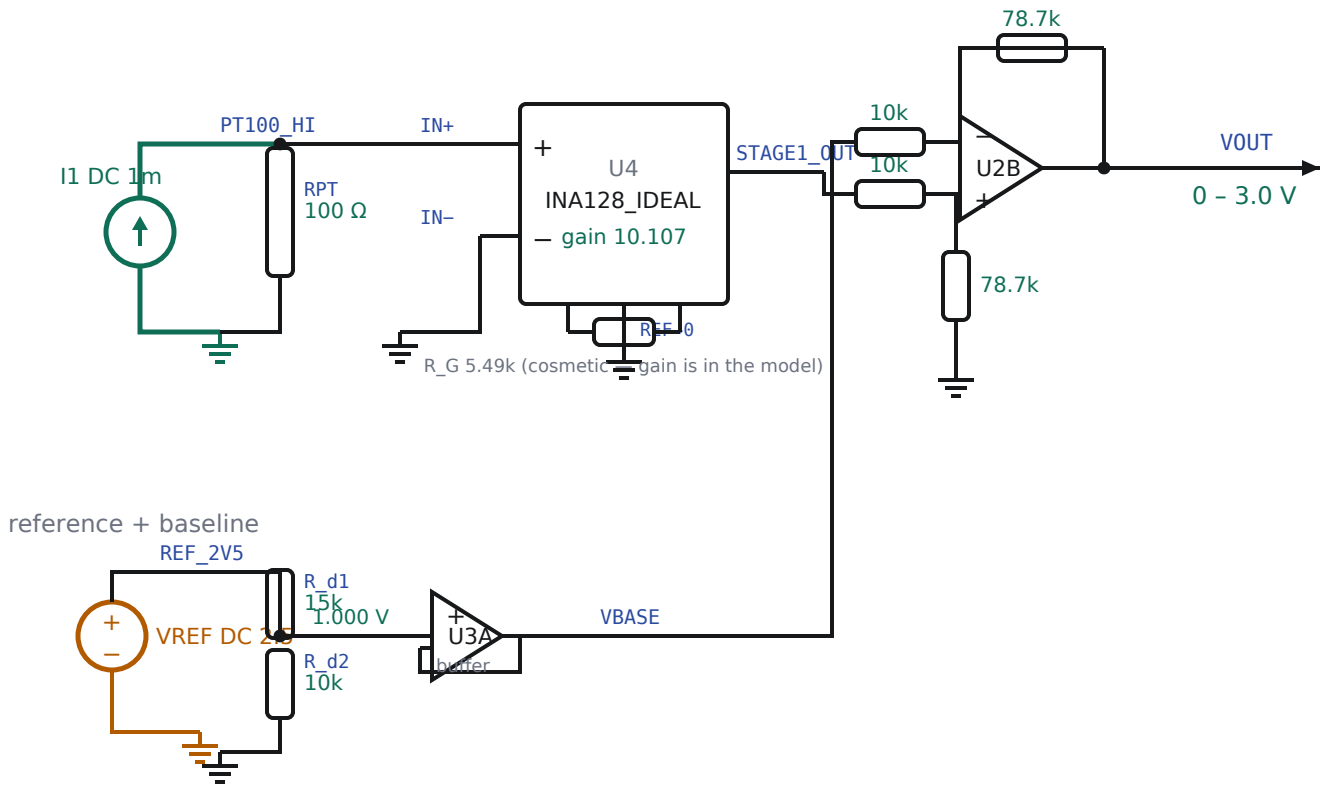
PT100 readout — simulation-ready

KiCad 8 / ngspice · ideal current source + ideal INA/op-amp macromodels · DC operating point + sweep

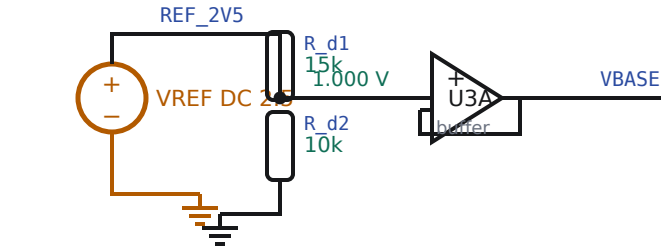
excitation (ideal source)

instrumentation amp (XU4)

difference amp (XU2B) · 0–3 V out



reference + baseline



supplies (rails via net labels +5V / –5V)



op-amp / INA V+ pins → +5V net,
V– pins → –5V net (net labels, not drawn as wires).

Ideal E-source models don't load the rails, but the
pins still need a driven net for a DC path.

place as schematic text (Place → Text):

```
.op  
.dc RPT 100 138.5 0.385
```

simulation setup

models — Simulation Model dialog:

→ “SPICE model from file:”

→ spice/ideal_models.lib, then Pin
Assignments tab to map pins.

U4 → INA128_IDEAL

OUT,IN+,IN–,REF,V+,V–,RG1,RG8

U2B, U3A → OPAMP_IDEAL

OUT,IN+,IN–,V+,V–

PT100 → plain R

refdes RPT, value 100 (not a model)

sources

I1 = DC 1m VREF = DC 2.5

V1 = DC 5 V2 = DC –5

ground

place a GND symbol = ngspice node 0

caps

omit decoupling for DC .op (open at DC)

note

Howland (U2A / R_set / R_H1–4) is

bypassed — ideal I1 stands in for it.

expected .op

I(RPT) = 1.000 mA

V(STAGE1_OUT) ≈ 1.0 V

V(VOUT) ≈ 0 V (→ 3.0 V at 100 °C)

PT100 readout — sim-ready

Portfolio Project 5

ngspice / KiCad 8

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DC .op / .dc